ALL INDIA COUNCIL FOR TECHNICAL EDUCATION (AICTE)

AICTE-Verify: Intelligent Cross-Document Scrutiny & Affiliation Risk Assessment System

Comprehensive Technical Architecture, End-to-End Operational Lifecycle, Machine Learning Forensic Pipeline Specifications, and Statutory Approval Process Handbook (APH 2024–25 to 2026–27) Integration Manual

Document Version: 2.4.0 (Production Release)	**Regulatory Base:** AICTE APH 2024-25 to 2026-27
Target Framework: Smart India Hackathon (SIH) 2024	**Classification:** Official Technical Architecture Manual
Engine Status: Multi-Model Microservices Operational	**Security Spec:** Zero-Trust SHA-256 Audit Trail

Executive Summary: AICTE-Verify is an automated regulatory scrutiny platform developed to resolve the acute inspection bottleneck across India's 10,000+ technical education institutions. Each academic year, institutions submit thousands of multipage regulatory artifacts (Land Title Deeds, Building Completion Form D, Fire Safety NOCs, Faculty Rosters, and Audited Financial Statements). AICTE-Verify combines multi-lingual optical character recognition (OCR), computer vision Error Level Analysis (ELA) for digital forgery detection, perceptual hashing (pHash) for cross-institutional certificate deduplication, an automated AICTE APH dimensional compliance engine, and a Cytoscape knowledge graph to detect ghost faculty syndicates and duplicated land parcels. The system enforces strict Role-Based Access Control (RBAC), ensuring institutions submit self-disclosures, Regional Officers adjudicate territorial jurisdictions, and Central Admins oversee national cryptographic hash chain integrity.

1. Complete Technology Stack & Architectural Topology

The system employs a decoupled, microservices-ready architecture comprising a high-performance reactive single-page frontend, an asynchronous REST API backend, and an intelligent forensic processing pipeline.

Architectural Tier	Core Technologies & Frameworks	Design Rationale & Production Purpose
Frontend Web Application	React 18.3, TypeScript 5.5, Vite 5.4, Tailwind CSS 3.4, React Router v6, Cytoscape.js 3.30, Axios	Strict type safety, zero-bundle compilation latency via Vite, atomic reactive UI state, hardware-accelerated force-directed graph visualization.
Backend API Engine	Python 3.12+, FastAPI 0.115, Uvicorn ASGI, SQLAlchemy 2.0 (AsyncIO), Pydantic v2 Settings & Schemas	High concurrency non-blocking I/O, automated OpenAPI/Swagger contract generation, strict request validation, native Python asyncio worker pools.
Database & Persistence	SQLite 3.45 with aiosqlite driver, Write-Ahead Logging (WAL) Mode, Relational Schema	ACID transactional compliance, zero-configuration local portability, seamless migration pathway to PostgreSQL/CockroachDB for distributed clustering.
Computer Vision & OCR	Tesseract 5.4 (Indic Multi-lingual: Eng/Tam/Hin), OpenCV 4.10, Pillow 10.4, NumPy 2.1	Local on-premise execution with zero cloud egress cost; adaptive thresholding, morphological skew correction, and localized pixel error analysis.

Digital Forensics Suite	Custom Error Level Analysis (ELA) Engine, DCT 64-bit Perceptual Hashing (pHash), Cryptographic X.509 ASN.1 Parser	Detects JPEG re-compression differentials (Photoshop copy-paste stamp tampering), cross-institution cloned certificates, and expired PKI signatures.
Large Language Model (LLM)	Google Gemini 1.5 Pro / Flash REST API Integration	Synthesizes multi-lingual statutory deficiency notices (English, Hindi, Tamil) conforming to administrative law standards and the AICTE APH Act.
Security & Cryptography	Argon2id Password Hasher, JWT Bearer (HS256/RS256), SHA-256 Merkle Audit Chain, PII Masking Engine	State-of-the-art memory-hard credential protection, stateless session scoping, tamper-evident immutable audit trails, and privacy protection for Aadhaar/PAN.

2. End-to-End System Workflow & Operational Lifecycle

The AICTE-Verify architecture models a deterministic, 9-stage pipeline that transitions an application from initial institutional self-disclosure to final officer adjudication and immutable ledger archiving.

Phase 1: Institutional Application & Self-Disclosure (/apply)

The college administrator selects the statutory category (Extension of Approval, Intake Increase, New Course, or New Institution) and inputs institutional coordinates, land dimensions, instructional built-up area, classrooms, CAD labs, faculty counts, and attaches required proofs under APH Annexure 2 (Affidavit 2, Fire NOC, Occupancy Form D, Satellite Map, Audited Balance Sheet).

Phase 2: Document Ingestion & SHA-256 Immutability Check

Upon upload, the backend computes the document's SHA-256 cryptographic checksum. A record is committed to the immutable audit log linking the upload to the acting user, IP address, and timestamp. File payloads are stored in an isolated filesystem vault.

Phase 3: Multi-Lingual OCR & Entity Extraction

The ingestion worker passes the document through OpenCV for grayscale transformation, Otsu adaptive thresholding, and deskewing. Tesseract 5 extracts textual blocks across English, Hindi, and Tamil scripts. LayoutLM and regex extractors parse key entities: Survey Numbers, Carpet Area in sq.m, Approved Intake, Faculty PAN hashes, and Issue/Expiry dates.

Phase 4: Automated Digital Forensic Analysis (ELA & pHash)

The forensics engine executes Error Level Analysis (ELA) by re-compressing images at a calibrated 95% JPEG quality level and computing absolute pixel error differentials. Digital alterations (e.g. Photoshop copy-pasted notary rubber stamps) produce concentrated error spikes. Concurrently, discrete cosine transform (DCT) 64-bit perceptual hashing compares the certificate against all certificates previously uploaded across all Indian institutions to detect duplicate reused files.

Phase 5: Simulated External Statutory Authority Gateway

The application query engine cross-references extracted identifiers against authoritative simulated external registries: (1) State Land Revenue Portals (Patta/Chitta / Bhulekh), (2) State Fire Services Headquarter NOC Portals, (3) Affiliating University Clearance Portals (Anna Univ, AKTU, SPPU), and (4) NSDL Income Tax PAN registries to detect multi-institutional employment.

Phase 6: AICTE Approval Process Handbook (APH) Norms Engine

The compliance engine applies statutory mathematical constraints: • Minimum Land Norms: Urban (≥5.0 Acres), Rural (≥7.5 Acres). • Infrastructure Ceiling: Intake ≤ (Built-up Area / 8.5 m²). • Faculty Cadre Ceiling: Intake ≤ (Verified Faculty × 20) with 1:2:6 cadre ratio. • CAD Terminal Norms: 1 workstation per 4 students in Engineering programmes. The system isolates the exact bottleneck constraining institutional capacity.

Phase 7: Composite Risk Scoring & Discrepancy Flag Generation

An algorithm weighs infractions across Land, Faculty, Forensics, and Regulatory dimensions to generate a Composite Risk Index (0 to 100). The application is triaged into Low Risk (eligible for auto-clearance), Medium Risk (manual audit), or High Risk (deficiencies).

Phase 8: Regional Officer Scrutiny & Automated Deficiency Notice Dispatch

The assigned Regional Officer inspects flagged anomalies via side-by-side split evidence viewers. For deficient applications, the officer generates an automated statutory deficiency letter synthesized by Gemini AI into formal administrative Hindi, Tamil, or English.

Phase 9: Cryptographic Hash Chain Verification & Collusion Graph Analysis

Every officer action, review note, and status change is hashed with the previous block's SHA-256 digest, preventing retrospective falsification. The Cytoscape network graph maps shared faculty PANs, identical land surveys, and overlapping educational trusts.

3. Detailed Deep Dive: AI & Machine Learning Models

AICTE-Verify incorporates six purpose-built artificial intelligence and algorithmic models. The section below details their technical architectures, operating mechanisms, and why each specific model was chosen for government scrutiny.

Model 1: Multi-Lingual OCR & Document Entity Extraction

Technology Core: Tesseract 5.4 OCR Engine + OpenCV Image Processing + LayoutLM Spatial Heuristic

Functional Task: Extract declared numerical and statutory values (Survey No, Land Area, Built-up sq.m, Faculty Names, Affiliation Ref, Dates) from scanned PDFs across English, Tamil, and Hindi.

Algorithmic Mechanism: Scanned documents are pre-processed using OpenCV: bilinear interpolation deskewing, Otsu adaptive thresholding for ink-bleed elimination, and morphological noise reduction. Tesseract's LSTM recurrent neural network decodes text glyphs into line blocks with bounding box coordinates. A rule-based spatial layout parser extracts key-value pairs.

Why This Specific Model: Government educational certificates across India are inherently multi-lingual (e.g. Tamil Nadu Land Pattas in Tamil script, Uttar Pradesh Revenue Khasras in Devanagari Hindi, and AICTE affidavits in English). Tesseract executes entirely on-premise without cloud latency, guarantees data privacy, and generates deterministic bounding box coordinates required as admissible legal evidence.

Model 2: Forensic Error Level Analysis (ELA) & Digital Tamper Detector

Technology Core: Error Level Analysis (ELA) Engine + JPEG Quantization Matrix Analysis

Functional Task: Detect digitally manipulated or copy-pasted rubber stamps, forged magistrate signatures, and digitally altered survey numbers on uploaded certificates.

Algorithmic Mechanism: When a digital document is edited (e.g., in Adobe Photoshop) and re-saved, the modified region is compressed at a different generation than the authentic background. The ELA engine re-saves the image at a calibrated 95% JPEG quality table and computes the pixel-by-pixel absolute error differential: E = |I_original - I_resaved| * ScaleFactor. Authentic paper scans exhibit uniform high-frequency noise, whereas forged notary seals display stark, glowing error residual clusters.

Why This Specific Model: Unscrupulous applicants often digitally copy authentic municipal seals from older approved documents and paste them onto expired certificates. ELA provides non-destructive, mathematically explainable forensic evidence that can be visually presented to scrutiny committees and hearing benches.

Model 3: Perceptual Hashing (pHash) for Cross-Institutional Document Cloning

Technology Core: Discrete Cosine Transform (DCT) 64-bit Perceptual Hash (pHash) Algorithm

Functional Task: Identify identical or visually cloned certificates submitted across different colleges in the same educational trust.

Algorithmic Mechanism: Unlike cryptographic hashes (SHA-256) where changing 1 bit completely changes the hash, pHash generates a fingerprint based on visual frequency distribution: (1) Resize image to 32x32 pixels, (2) Convert to grayscale, (3) Compute Discrete Cosine Transform (DCT) matrix, (4) Retain low frequencies (8x8 top-left matrix), (5) Calculate median frequency, (6) Generate 64-bit binary hash. When comparing two documents, the Hamming distance D_H is computed. If D_H <= 5 bits, the certificates are mathematically verified to be duplicates.

Why This Specific Model: Educational trusts running multiple polytechnics and engineering colleges often upload the exact same Fire Safety NOC or Building Plan for multiple campuses. Cryptographic hashing fails if the scan has different compression or DPI. pHash detects visual structural equivalence regardless of minor scan variations.

Model 4: Approval Process Handbook (APH) Statutory Norms Engine

Technology Core: Constrained Optimization Rule Engine & Bottleneck Resolution Algorithm

Functional Task: Enforce dimensional statutory norms from AICTE APH Chapter I & VII, calculate infrastructure and faculty ceilings, and identify limiting bottlenecks.

Algorithmic Mechanism: The engine models statutory constraints as a multi-variable linear optimization problem: C_infra = floor(BuiltUpArea / 8.5) students; C_faculty = FacultyCount * 20 students; ApprovedIntake = min(AppliedIntake, C_infra, C_faculty). If C_faculty < C_infra, the engine flags a 'Faculty Bottleneck' and calculates the exact staff shortage.

Why This Specific Model: Statutory regulatory compliance cannot rely on black-box heuristics. AICTE regulations require unambiguous, legally auditable calculations. The norms engine provides deterministic mathematical explanations that eliminate subjective scrutiny bias.

Model 5: Cytoscape Knowledge Graph & Cross-College Collusion Network

Technology Core: Cytoscape.js Force-Directed Graph Engine + Multi-Entity Bipartite Resolution

Functional Task: Detect ghost faculty registered on multiple payrolls simultaneously across different states and identify overlapping land survey numbers.

Algorithmic Mechanism: Entities (Institutions, Faculty PANs, Land Parcels, Educational Trusts) are modeled as nodes V in a directional multi-graph G = (V, E). Edges E represent relationships (employed_at, owns_land, managed_by). The graph engine runs anomaly traversal queries: if a faculty node has out-degree >= 2 with 'employed_at' edges to colleges in different states (e.g. Dr. Ramanathan listed in both Chennai and Pune), an anomalous clique is flagged.

Why This Specific Model: Traditional relational database queries struggle with multi-hop collusion patterns across disparate regional tables. Graph topology provides instant visual and algorithmic identification of fraudulent institutional syndicates.

Model 6: Generative Large Language Model (Gemini AI) for Statutory Notices

Technology Core: Google Gemini 1.5 Pro / Flash Multi-Lingual Generative Architecture

Functional Task: Synthesize observed technical infractions into legally sound, formal administrative Deficiency Hearing Notices in English, Hindi, and Tamil.

Algorithmic Mechanism: The LLM receives structured JSON payload summaries of observed forensic and regulatory flags. Using strict system prompts encoding AICTE administrative law principles, it synthesizes formal deficiency notices quoting exact APH clauses, granting a 7-day statutory hearing window, and detailing remedial requirements.

Why This Specific Model: Administrative natural justice requires that institutions receive clear, articulate deficiency notices rather than raw database error codes. Gemini provides high-fidelity, legally formatted multi-lingual drafting, ensuring colleges in Tamil Nadu or Uttar Pradesh receive notices in their regional language.

4. Role-Based Access Control (RBAC) & Governance Matrix

To preserve administrative integrity and statutory natural justice, AICTE-Verify enforces strict role-based data isolation. Neither the Central Admin nor Regional Officers possess document upload capabilities. Uploads are strictly restricted to applicant institutions.

System Persona	Territorial Jurisdiction	Document Upload Permission	Permitted Scrutiny & Governance Capabilities
College User (college1, college2, etc.)	Isolated to single assigned institution	**YES (Exclusive)** Uploads Annexure 1 & 2 files	Submit new EoA applications (/apply), declare metrics, upload statutory PDFs, view scrutiny findings, read deficiency notices, download official orders.
Regional Officer (RO) (officer1 South, officer2 North)	Assigned geographical territory (e.g. Southern = TN/AP, Northern = UP/UK)	**NO ACCESS** Zero upload capability	Scrutinize regional applications, inspect OCR text and ELA tamper heatmaps, evaluate APH compliance ratios, adjudicate review queue, issue bilingual deficiency notices, grant approvals.
Central Administrator (admin)	National Apex (All 8 AICTE Regions across India)	**NO ACCESS** Zero upload capability	National scrutiny oversight, SHA-256 cryptographic audit ledger verification, simulate tamper detection, monitor ML model accuracy metrics, analyze cross-state collusion graphs, manage RO officer accounts.

5. AICTE Approval Process Handbook (APH 2024-27) Statutory Norms

The system directly implements the legal and dimensional standards established in the AICTE Approval Process Handbook:

- **Land Contiguity & Area (Appendix 4.1):** Engineering institutions in Rural areas require minimum 7.5 contiguous acres; Urban areas require minimum 5.0 contiguous acres; Mega cities require 2.5 contiguous acres.
- **Instructional Built-Up Area (Appendix 4.2):** Classrooms, laboratories, drawing halls, and workshops must satisfy minimum 8.5 m^2 per student of total sanctioned intake.
- **Faculty-to-Student Cadre Ratio (Clause 7.2):** Undergraduate Engineering programmes mandate a strict 1:20 teacher-student ratio, with a cadre distribution of 1 Professor : 2 Associate Professors : 6 Assistant Professors.
- **Computing & CAD Workstation Facilities (Clause 5.4):** Minimum 1 computer workstation per 4 students in Engineering & Technology, connected via high-speed LAN with licensed design and simulation software.
- **Mandatory Fire Safety Certification (Clause 3.1):** Annual Fire Safety NOC issued by the state Fire Disaster Management Command is a non-derogable condition precedent for grant of Extension of Approval (EoA).

6. Project Directory Pathways & File Reference Guide

The complete AICTE-Verify codebase is structured cleanly within the active workspace. Key absolute file pathways are detailed below:

Component / Deliverable	Absolute File System Pathway
Technical Architecture PDF	AICTE_Verify_System_Architecture_And_Technical_Documentation.pdf (Project Root)
Frontend Root Directory	c:\Users\JANAV KARTHICK\Documents\SIH 2\aicte-verify\frontend
Backend Root Directory	c:\Users\JANAV KARTHICK\Documents\SIH 2\aicte-verify\backend
New EoA Application Portal	aicte-verify\frontend\src\pages\NewApplicationPage.tsx
Application Detail & Scrutiny	aicte-verify\frontend\src\pages\ApplicationDetailPage.tsx

Collusion Graph (Cytoscape)	aicte-verify\frontend\src\pages\GraphPage.tsx
Deficiency Notice Generator	aicte-verify\frontend\src\pages\LettersPage.tsx
Applications API Router	aicte-verify\backend\app\api\applications.py
Document Forensics Pipeline	aicte-verify\backend\app\services\forensics\service.py
Cryptographic Audit Ledger	aicte-verify\backend\app\services\audit\service.py
Database SQLite File	aicte-verify\backend\aicte_verify.db

Certification: This technical architecture manual has been compiled and verified against the live execution state of the AICTE-Verify system. All microservices, machine learning modules, and cryptographic hash chain algorithms comply with the statutory mandates of the All India Council for Technical Education.